

Ligation sequencing DNA V14 (SQK-LSK114)

Overview

Before beginning, we strongly recommend that first-time users read and review the complete online protocol on the Nanopore Community. The online protocol provides the full instructions, additional commentary, and considerations for various steps, as well as guidance on input amounts, DNA handling, and sequencing. **This reference card is designed for experienced users.**

1. DNA repair and end-prep (~35 minutes)

- 1 Thaw DNA Control Sample (DCS) at room temperature, spin down, mix by pipetting, and place on ice.
- 2 Prepare the NEB reagents in accordance with manufacturer's instructions, and place on ice. For optimal performance, NEB recommend the following:
 1. Thaw all reagents on ice.
 2. Flick and/or invert the reagent tubes to ensure they are well mixed.
Note: Do not vortex the FFPE DNA Repair Mix or Ultra II End Prep Enzyme Mix.
 3. Always spin down tubes before opening for the first time each day.
 4. Vortex the FFPE DNA Repair Buffer v2 to ensure it is well mixed.
Note: This buffer may contain a white precipitate. If this occurs, allow the mixture to come to room temperature and pipette the buffer several times to break up the precipitate, followed by a quick vortex to mix.
 5. The FFPE DNA Repair Buffer v2 may have a yellow tinge and is fine to use if yellow.
- 3 Prepare the DNA in nuclease-free water:
 1. Transfer 1 µg (or 100-200 fmol) input DNA into a 1.5 ml Eppendorf DNA LoBind tube.
 2. Adjust the volume to 47 µl with nuclease-free water.
 3. Mix thoroughly by pipetting up and down, or by flicking the tube.
 4. Spin down briefly in a microfuge.
- 4 In a 0.2 ml thin-walled PCR tube, mix the following:

Between each addition, pipette mix 10-20 times.

Reagent	Volume
DNA from the previous step	47 µl
DNA CS (optional)	1 µl
NEBNext FFPE DNA Repair Buffer v2	7 µl
NEBNext FFPE DNA Repair Mix	2 µl
Ultra II End-prep Enzyme Mix	3 µl
Total	60 µl

- 5 Thoroughly mix the reaction by gently pipetting and briefly spinning down.
- 6 Using a thermal cycler, incubate at 20°C for 5 minutes and 65°C for 5 minutes. Then cool down to between 4°C and 20°C on the thermal cycler or place the samples on ice.
- 7 Resuspend the AMPure XP Beads (AXP) by vortexing.
- 8 Spin down and transfer the DNA sample to a clean 1.5 ml Eppendorf DNA LoBind tube.
- 9 Add 60 µl of resuspended the AMPure XP Beads (AXP) to the end-prep reaction and mix by flicking the tube.
- 10 Incubate on a Hula mixer (rotator mixer) for 5 minutes at room temperature.

- 11 Freshly prepare 500 µl of 80% ethanol in nuclease-free water.
- 12 Spin down the sample and pellet on a magnet until supernatant is clear and colourless. Keep the tube on the magnet, and pipette off the supernatant.
- 13 Keep the tube on the magnet and wash the beads with 200 µl of freshly prepared 80% ethanol without disturbing the pellet. Remove the ethanol using a pipette and discard.
- 14 Repeat the previous step.
- 15 Spin down and place the tube back on the magnet. Pipette off any residual ethanol. Allow to dry for ~30 seconds, but do not dry the pellet to the point of cracking.
- 16 Remove the tube from the magnetic rack and resuspend the pellet in 61 µl nuclease-free water by gently pipetting up and down or by flicking the tube. Incubate for 2 minutes at room temperature.
- 17 Pellet the beads on a magnet until the eluate is clear and colourless, for at least 1 minute.
- 18 Remove and retain 61 µl of eluate into a clean 1.5 ml Eppendorf DNA LoBind tube.

2. Adapter ligation and clean-up (~20 minutes)

- 1 Spin down the Ligation Adapter (LA) and Salt-T4® DNA Ligase, and place on ice.
- 2 Thaw Ligation Buffer (LNB) at room temperature, spin down and mix by pipetting. Due to viscosity, vortexing this buffer is ineffective. Place on ice immediately after thawing and mixing.
- 3 Thaw the Elution Buffer (EB) at room temperature and mix by vortexing. Then spin down and place on ice.
- 4 Thaw either Long Fragment Buffer (LFB) or Short Fragment Buffer (SFB) at room temperature and mix by vortexing. Then spin down and keep at room temperature.
- 5 In a 1.5 ml Eppendorf DNA LoBind tube, mix in the following order:
Between each addition, pipette mix 10-20 times.

Reagent	Volume
DNA sample from the previous step	60 µl

Reagent	Volume
Ligation Adapter (LA)	5 µl
Ligation Buffer (LNB)	25 µl
Salt-T4® DNA Ligase	10 µl
Total	100 µl

- 6 Thoroughly mix the reaction by gently pipetting and briefly spinning down.
- 7 Incubate the reaction for 10 minutes at room temperature.
- 8 Resuspend the AMPure XP Beads (AXP) by vortexing.
- 9 Add 40 µl of resuspended AMPure XP Beads (AXP) to the reaction and mix by flicking the tube.
- 10 Incubate on a Hula mixer (rotator mixer) for 5 minutes at room temperature.
- 11 Spin down the sample and pellet on a magnet. Keep the tube on the magnet, and pipette off the supernatant when clear and colourless.
- 12 Wash the beads by adding either 250 µl Long Fragment Buffer (LFB) or 250 µl Short Fragment Buffer (SFB). Flick the beads to resuspend, spin down, then return the tube to the magnetic rack and allow the beads to pellet. Remove the supernatant using a pipette and discard.
- 13 Repeat the previous step.
- 14 Spin down and place the tube back on the magnet. Pipette off any residual supernatant.

Allow to dry for ~30 seconds, but do not dry the pellet to the point of cracking.

- 15 Remove the tube from the magnetic rack and resuspend the pellet in 15 µl Elution Buffer (EB). Spin down and incubate for 10 minutes at room temperature. For high molecular weight DNA, incubating at 37°C can improve the recovery of long fragments.
 - 16 Pellet the beads on a magnet until the eluate is clear and colourless, for at least 1 minute.
 - 17 Remove and retain 15 µl of eluate containing the DNA library into a clean 1.5 ml Eppendorf DNA LoBind tube.
- Dispose of the pelleted beads
- 18 Depending on your DNA library fragment size, prepare your final library in 12 µl of Elution Buffer (EB).

Fragment library length	Flow cell loading amount
Very short (<1 kb)	100 fmol
Short (1-10 kb)	35-50 fmol
Long (>10 kb)	300 ng

Note: If the library yields are below the input recommendations, load the entire library.

If required, we recommend using a mass to mol calculator such as the [NEB calculator](#).

3. Priming and loading the MinION and GridION Flow Cell (~5 minutes)

- 1 Thaw the Sequencing Buffer (SB), Library Beads (LIB) or Library Solution (LIS, if using), Flow Cell Tether (FCT) and Flow Cell Flush (FCF) at room temperature before mixing by vortexing. Then spin down and store on ice.
 - 2 To prepare the flow cell priming mix with BSA, combine Flow Cell Flush (FCF) and Flow Cell Tether (FCT), as directed below. Mix by pipetting at room temperature.
- In a suitable tube for the number of flow cells, combine the following reagents:

Reagent	Volume per flow cell
Flow Cell Flush (FCF)	1,170 µl
Bovine Serum Albumin (BSA) at 50 mg/ml	5 µl
Flow Cell Tether (FCT)	30 µl
Total volume	1,205 µl

- 3 Open the MinION or GridION device lid and slide the flow cell under the clip. Press down firmly on the priming port cover to ensure correct thermal and electrical contact.
- 4 Slide the flow cell priming port cover clockwise to open the priming port.
- 5 After opening the priming port, check for a small air bubble under the cover. Draw back a small volume to remove any bubbles:
 1. Set a P1000 pipette to 200 µl
 2. Insert the tip into the priming port
 3. Turn the wheel until the dial shows 220-230 µl, to draw back 20-30 µl, or until you can see a small volume of buffer entering the pipette tip

Note: Visually check that there is continuous buffer from the priming port across the sensor array.

- 6 Load 800 µl of the priming mix into the flow cell via the priming port, avoiding the introduction of air bubbles. Wait for five minutes. During this time, prepare the library for loading by following the steps below.
- 7 Thoroughly mix the contents of the Library Beads (LIB) by pipetting.
- 8 In a new 1.5 ml Eppendorf DNA LoBind tube, prepare the library for loading as follows:

Reagent	Volume per flow cell
Sequencing Buffer (SB)	37.5 µl
Library Beads (LIB) mixed immediately before use, or Library Solution (LIS), if using	25.5 µl
DNA library	12 µl
Total	75 µl

- 9 Complete the flow cell priming:
 1. Gently lift the SpotON sample port cover to make the SpotON sample port accessible.
 2. Load **200 µl** of the priming mix into the flow cell priming port (**not** the SpotON sample port), avoiding the introduction of air bubbles.
- 10 Mix the prepared library gently by pipetting up and down just prior to loading.
- 11 Add 75 µl of the prepared library to the flow cell via the SpotON sample port in a dropwise fashion. Ensure each drop flows into the port before adding the next.
- 12 Gently replace the SpotON sample port cover, making sure the bung enters the SpotON port and close the priming port.
- 13 Place the light shield onto the flow cell, as follows:
 1. Carefully place the leading edge of the light shield against the clip.

Note: Do not force the light shield underneath the clip.
 2. Gently lower the light shield onto the flow cell. The light shield should sit around the SpotON cover, covering the entire top section of the flow cell.

4. Flow cell reuse and returns

- 1 After your sequencing experiment is complete, if you would like to reuse the flow cell, please follow the Flow Cell Wash Kit protocol and store the washed flow cell at +2°C to +8°C.

The [Flow Cell Wash Kit protocol](#) is available on the Nanopore Community.

- 2 Alternatively, follow the returns procedure to send the flow cell back to Oxford Nanopore. Instructions for returning flow cells can be found [here](#).